SCIENCE 1.6 **Name:** ***ANSWERS***

WORKSHEET SIX **TYPES AND TRANSFER OF ENERGY**

$E_p = mgh$ $E_k = \frac{1}{2}mv^2$

1. What is the unit for energy? ***Joules*** What is the symbol for this unit? ***J***
2. List the names of six types of energy. ***Heat, Kinetic, Gravitational potential, Solar, Sound, Elastic potential, Chemical potential.***
3. Name the main type(s) of energy present in the following situations.
   (a) An unlit match. ***Chemical potential energy***
   (b) A stretched rubber band ***Elastic potential energy***
   (c) An orbiting satellite ***Kinetic energy, Gravitational potential energy***
   (d) A thunder clap ***Sound energy***
   (e) A bolt of lightning ***Electrical energy and light energy***
   (f) A kicked soccer ball ***Kinetic energy***
4. The sun is a giant hydrogen bomb. It uses ***nuclear*** energy to create solar energy. This solar energy is a mixture of different types of radiant energy such as ***light*** and ***heat***. This radiant energy can travel through ***space*** to earth and the earth absorbs some of it and ***reflects*** some of it. Plants use chlorophyll to absorb light energy except for the colour ***green*** which they reflect. They store this energy in their leaves and roots as ***chemical*** potential energy.
5. Write energy transfer equations for the following situations.
   (a) An electric bell is switched on.
   Electrical energy → Kinetic energy → Sound energy
   (b) A bullet is fired up into the air.
   Chemical potential energy → Kinetic energy → Gravitational PE
   (c) A pine-cone falls off the top of a pine tree.
   Gravitational potential energy → Kinetic energy
6. Usually when energy is transferred from one type to ***another***, some of the energy is also turned into ***Heat***. In a car motor, less than 50% of the chemical

energy stored in the petrol is turned into the ***kinetic*** energy of the moving car. The car engine gets very ***hot*** and this unusable heat must be removed. The car radiator is filled with ***water*** which circulates through the motor and removes the heat. The hot water then circulates through the radiator and is ***cooled*** by the air passing over it.

7. In each of the following pairs, one member of the pair has more energy. Which is it?
(a) A kg of lead and a kg of feathers on the same shelf. ***Same***
(b) A 50cent piece on a table or a 50cent piece on the ground beside it? ***Table***
(c) A cricket-ball hit by Noddy or a cricketball hit by Superman?? ***Superman***

8. Calculate the gravitational potential energy in a 1.6 kg pot-plant which is on a shelf 1.4 m above the floor. (Use g = 10 Nkg^{-1})

E = mgh = 1.6 x 10 x 1.4 = 22.4 J

9. Calculate the gravitational potential energy in a 0.4kg seagull which is flying at a height of 200 m above the ground. (Use g = 10 Nkg^{-1})

E = mgh = 0.4 x 10 x 200 = 800 J

10. Calculate what height a 3kg object must have above the ground if it has 540J of gravitational potential energy? (Use g = 10 Nkg^{-1})

E = mgh 540 = 3 x 10 x h h = 18 m

11. A boy and his bike have a combined mass of 70 kg. If the boy is cycling at 15ms^{-1} how much kinetic energy is present?

$E = \frac{1}{2}mv^2 = \frac{1}{2} \times 70 \times 15^2 = 7875$ J

12. A bullet with mass 15g is fired at 200ms^{-1}. How much kinetic energy has it got?

$E = \frac{1}{2}mv^2 = \frac{1}{2} \times 0.015 \times 200^2 = 300$ J

13. A 200g orange falls off a tree and just before it hits the ground it reaches a speed of 10 ms^{-1}.
(a) Calculate how much kinetic energy it has.

$E = \frac{1}{2}mv^2 = \frac{1}{2} \times 0.2 \times 10^2 = 10$ J

(b) Calculate how high it was growing on the tree before it fell off. (HARD!)

E = mgh 10 = 0.2 x 10 x h h = 5 m